

Thomas Bentivoglio

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EDUCATION

Case Western Reserve University

May 2027

Bachelor of Science Computer Science and Mechanical Engineering

- Minor in mathematics

Relevant Courses:

- Full-Stack, Data-structures, Algorithms, Discrete Math, Algorithms, Linear Algebra, Software Engineering, Programming Language Concepts, Databases
- Mechanical Manufacturing, Statics and Strength of Materials, Differential Equations, Thermodynamics, Fluid Mechanics, Design and Manufacturing, Heat transfer

SKILLS/PROJECTS

- Programming: Java, Python, Java Script
 - Made the game 2048 in Java and JavaScript
 - Developed a web-app with a team that allows students to find free food
- Full Stack Development (HTML, CSS, Databases, React, MUI, etc.)
- MatLab
- SolidWorks (CSWA Certified, working towards professional)
 - Designed a water jug lift for elderly people with a team
- Certified in using lathe, mill, basics in CNC machine and water jet
 - Trained in Think[box], demonstrated skills by machining a hammer

RESEARCH EXPERIENCE

Cable-Driven Hyper redundant Robot Research (CyPhiLab)

March 2025 – Present

- Developing and testing control strategies for a cable-driven, hyper-redundant robot (“SpiRob”) using Python, ROS2, and MuJoCo simulation
- Integrated CAD models from Onshape into MuJoCo for physics-based simulation, defining tendon actuation and joint constraints
- Implemented trajectory generation and Kalman filter-based state estimation for accurate robot control
- Collaborated with faculty and graduate students on robotic manipulation and human-machine interaction experiments

LEADERSHIP/ACTIVITIES

CWRUbotix (Case Western Reserve University Robotics Club)

September 2024 – Present

- MateROV(Submarine robot): Mechanical team and software team
 - Design manipulators and programming the robots autonomous functions using python
- Combat Robotics: Mechanical team
 - Manufacture and design with rapid prototyping using 3D printers

Autonomous Snow Plow Team

October 2025 – Present

- Engineered cost-map representations for autonomous operations and obstacle avoidance
- Developed waypoint-based path planning algorithms used by a firstplace snow plow robot